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Snehith Nayak

SF Bay Area, CA

EDUCATION

University of Michigan Ann Arbor | *M.S. Applied Data Science* Aug 2025 - Present
University of California Santa Barbara | *B.S. Computer Engineering* Jun 2024

WORK EXPERIENCE

Applied Materials | *Software Engineer AIx* | Distributed Systems, ML Infra Santa Clara, CA | July 2024 - Present

- Engineered a real-time data platform on Kubernetes, Apache Kafka (AWS MSK), and ClickHouse/DuckDB, ingesting ~35M rows/month of fab metrology and process data into a multi-billion-row store powering search, analytics, and recommendations across process-development experiments.
- Took the platform 0→1 to production (Feb 2025) and scaled to 600+ monthly active users, hardening the pipeline with priority-tiered Kafka topics, dead-letter queues, and circuit breakers for fault isolation.
- Built a recommendation layer that analyzes linked experiment and metrology data to suggest which process parameters engineers should tune, providing data-driven starting points for new semiconductor process recipes.
- Designed an experiment-tracking and metadata-logging system that makes every fab process and metrology run reproducible and directly comparable by configuration and result.

KLA | *Machine Learning Engineer Intern - BBP* | Python, Image Classification Milpitas, CA | June-Sept 2023

- Developed a machine-learning autoencoder using TensorFlow for BBP Nuisance Filtering, reducing cluster count by 30% in IBM CAD and Samsung GMK dataset compared to DBG2 (current implementation).
- Engineered an advanced, adaptable framework using convolutional layers, optimizing defect detection and classification across multiple datasets, specialized by training with unlabeled BBP hotspot images.
- Utilized HDBSCAN for efficient clustering of latent features, validated by silhouette scores.

Synopsys | *Application Development Intern* | Java, SQL, Data Processing Santa Clara, CA | June-Sept 2022

- Automated contractor onboarding process saving 2,200 employee hours annually, improving system efficiency for over 2,000 users.

PROJECTS

Pixel Pane – AI Assistant | SwiftUI, MLX, Cloudflare Workers, Anthropic API May 2026

- Built model-agnostic SwiftUI macOS AI assistant supporting any MLX-compatible backend, Anthropic Claude, and Apple Foundation Models — with per-provider health detection and lifecycle management.
- Deployed Cloudflare Workers proxy normalizing Anthropic streaming into SSE with per-user rate limiting, KV quota enforcement, and cryptographic request tracing.
- Engineered extensible action dispatch system with heuristic content classification, built-in handlers for translation and code debugging, and user-defined side effects.
- Designed permission-gated security model with explicit file grants and scoped side-effect authorization before any system-level operation.

iso-code – Safe Worktree Management for AI Agents | Rust, CLI, MCP Server, Git Apr 2026

- Built a Rust library, CLI, and MCP server providing safe git worktree lifecycle management for AI coding agents across Claude Code, Cursor, and Copilot.
- Prevented data loss from unmerged-commit deletion, orphaned worktrees, and git-crypt corruption with atomic state writes and cleanup-on-failure guarantees.

Health Monitoring Wearable – Capstone | Python, C, Java, ASIC Design Nov 2023-June 2024

- Designed and manufactured a specialized health monitoring wearable for nursing home residents, focusing on comprehensive remote monitoring and enhanced data accuracy.
- Incorporated advanced sensors in the wearable for vital signs, activity tracking, temperature monitoring, ambient noise detection, and continuous heart rate and oxygen level measurement.
- Engineered a data-efficient Android app for real-time health data visualization and trend analysis.
- Designed and manufactured a 4-Layer PCB design in Altium.

SKILLS

Languages: Python, SQL, C++, C
ML & Data: PyTorch, TensorFlow, ClickHouse, DuckDB
Infrastructure: Kubernetes, Kafka, AWS (MSK, S3, CloudWatch), Docker, FastAPI, Git
AI tooling: MCP, MLX